Drawing and Using Informal Lines of Fit

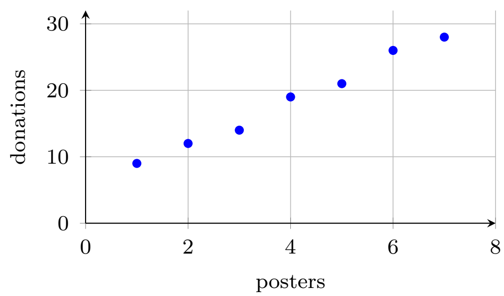
Drawing a line of fit, finding its slope, writing its equation, and predicting with it

Directions

- A **line of fit** is a single straight line drawn through the middle of a scatter plot, with about as many points above it as below. It does not have to touch any point.
- Slope of a line through (x_1, y_1) and (x_2, y_2): $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. Equation: $y = mx + b$, where b is the value where the line crosses the vertical axis.
- In Part A each problem has its own scatter plot. In Part B you are given the line and work from its equation.

Part A — working on the scatter plot

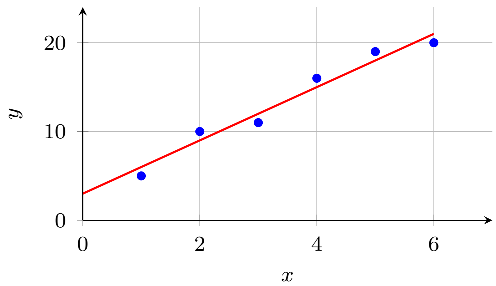
1. Each point shows a week of a fundraiser: posters put up and donations received.



(a) The greatest number of donations plotted is __________

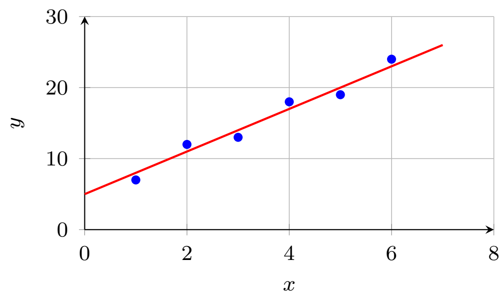
(b) Use a ruler to draw a line of fit on the plot above.

2. A line of fit has already been drawn. It passes exactly through $(1, 6)$ and $(5, 18)$. Find its slope.



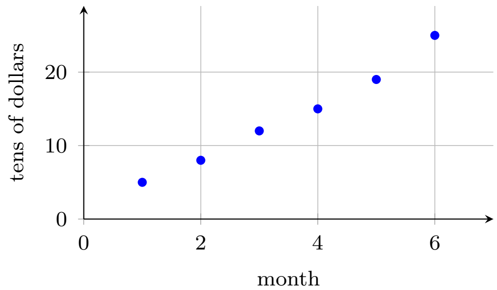
Answer: ____________________

3. The line of fit drawn below passes exactly through $(0, 5)$ and $(4, 17)$.



(a) Slope of the line = __________ (b) Value the line predicts at $x = 7$ is __________

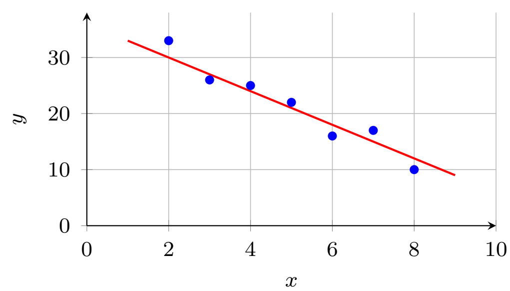
4. Each point shows a month of a savings plan: months elapsed and dollars saved, in tens.



(a) The mean of the six plotted values is _________

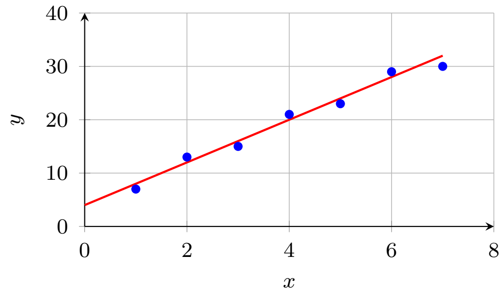
(b) Draw a line of fit that passes through the middle of the data.

5. This line of fit passes exactly through $(2, 30)$ and $(8, 12)$. Find its slope.



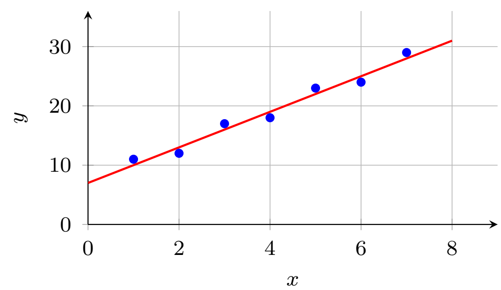
Answer: __________________

6. The line of fit drawn below passes exactly through $(1, 8)$ and $(6, 28)$, and it crosses the vertical axis at 4.



(a) Slope = ________ (b) Value predicted at $x = 10$ is ________

7. The line of fit drawn below passes exactly through $(0, 7)$ and $(5, 22)$.



(a) Slope = ________ (b) Equation of the line: $y =$ ________

(c) The value the equation gives at $x = 8$ is ________

Part B — working from the equation

8. For a class, the line of fit relating hours of revision x to test score y is $y = 6x + 48$. Predict the score of a student who revises for 7 hours.

Answer: ________________

9. Use the same line, $y = 6x + 48$.

(a) The score it predicts for 4 hours of revision is ________

(b) Explain what the slope 6 means for a student in this class.

____

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10. A line of fit passes exactly through $(2, 26)$ and $(6, 14)$.

(a) Slope = ________ (b) Equation: $y =$ ________

(c) The value the line predicts at $x = 9$ is ________

11. A class measured the height of a bean plant, in centimetres, on each of days 1 to 10. Their line of fit is $y = 2.5x + 12$.

(a) The height it predicts on day 6 is ________

(b) Explain why using this line to predict the height on day 40 would not be trustworthy.

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12. A class recorded minutes of daily exercise x and resting pulse y in beats per minute for every student. Their line of fit is $y = -0.4x + 78$.

(a) The pulse it predicts for 25 minutes of exercise is ________

(b) The change in predicted pulse for each extra 10 minutes of exercise is ________

(c) Explain whether this line shows that exercising more *causes* a lower resting pulse.

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